

High-Level Requirements

Functional Requirements

FR1: The system shall allow a cashier to process a sales transaction by scanning the product barcode and calculating the total automatically.

FR2: The system shall update inventory automatically after each completed sale.

FR3: The system shall allow managers/administrators to add, edit, and remove (including discontinued) products.

FR4: The system shall generate daily, weekly, and monthly sales reports.

FR5: The system shall manage user accounts and support role-based access for Admin, Manager, and Cashier roles.

FR6: The system shall allow users to search for products by name, category, or barcode.

FR7: The system shall be able to communicate with external systems (e.g. payment gateway, SMS, and email services) to support online payment and notifications.

FR8: The system shall allow administrators to create, disable, or remove employee (cashier) user accounts.

FR9: The system shall allow users to log in using a username and password.

FR10: The system shall allow administrators to view daily, weekly, and monthly sales reports.

FR11: The system shall allow administrators to view inventory reports showing current stock levels.

FR12: The system shall allow administrators to export reports for business analysis.

FR13: The system shall allow users to view product details, including price and stock availability.

FR14: The system shall allow the cashier to process payments, including online/digital payment methods.

FR15: The system shall allow the cashier to apply authorized discounts during checkout.

Non-Functional Requirements

NFR1: The system shall process a sales transaction within 2 seconds.

NFR2: The system shall ensure data security through authentication and role-based access control.

NFR3: The system shall be user-friendly and require minimal training to learn.

NFR4: The system shall maintain data integrity and prevent data loss, including during system interruptions.

NFR5: The system shall be scalable to support future expansion (e.g. additional store branches).

NFR6: The system shall work both online (to send reports and stock summaries to the cloud) and offline (continuing to process sales during network interruptions).
